Exercise – Understanding convergence concepts using simulation

1. Aim of the exercise

This exercise illustrates convergence concepts that underlie many econometric derivations. Using simulation, we study how estimation error evolves as the sample size increases and how this behavior can be formalized through stochastic convergence, small-o_p notation, and Big-$\mathcal{O}_p$ notation.

2. Theoretical background

A central question in econometrics is how estimation error behaves as the sample size increases. The key tools used to study this question are stochastic convergence, the Law of Large Numbers (LLN), and the Central Limit Theorem (CLT). These concepts provide the theoretical foundation for the simulations and convergence results developed in this exercise. Readers are advised to first review the dedicated LLN and CLT exercises, where these results are introduced and discussed in greater detail.

We first recall the deterministic notion of convergence. A sequence of numbers $\{x_n\}$ converges to a constant μ if, for any $\varepsilon > 0$, there exists an N such that

$$|x_n - \mu| < \varepsilon$$

for all $n > N$.

When the sequence is random, this condition cannot generally be guaranteed for every realization. Instead, we require that the probability of being within ε of the limit approaches one as the sample size increases.

A sequence of random variables $\{x_n\}$ converges in probability to a constant μ if, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|x_n - \mu| < \varepsilon) = 1.$$

This concept, also known as stochastic convergence, states that the probability of being arbitrarily close to μ approaches one as the sample size grows.

The Weak Law of Large Numbers provides a fundamental application of convergence in probability by establishing that the sample mean converges to the population mean.

Let

$$\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

where x_i are i.i.d. with mean μ. The Weak Law of Large Numbers states that

$$\lim_{n\to\infty} \mathbb{P}(|\bar{x}_n - \mu| < \varepsilon) = 1.$$

That is, the sample mean converges in probability to the population mean.

While the LLN establishes convergence, the CLT characterizes the remaining estimation error. In particular,

$$\bar{x}_n \overset{a}{\sim} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right),$$

or equivalently,

$$\sqrt{n}\left(\bar{x}_n - \mu\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\right).$$

In practice, researchers often approximate the distribution of the sample mean as $\bar{x}_n \sim \mathcal{N}(\mu, \sigma^2/n)$. However, as the sample size n grows to infinity, this variance shrinks to zero, and the distribution collapses to a single point. Multiplying the estimation error $(\bar{x}_n - \mu)$ by $\sqrt{n}$ counteracts this shrinkage. It stabilizes the variance at σ^2, which allows us to find a meaningful limiting probability distribution.

It is not clear from the theorem alone how rapidly the estimation error decreases as the sample size grows. One of the main goals of this exercise is to develop intuition for this relationship through simulation and connect it to the convergence concepts introduced in following sections.

3. Average absolute error and convergence rate

In the dedicated exercise on the Law of Large Numbers, we used a coin-toss experiment to demonstrate how the sample mean approaches the population mean as the sample size increases. In this exercise, we return to the same example and use it as a running illustration of the convergence concepts introduced below.

To investigate how rapidly the sample mean converges to the population mean, we calculate the average absolute error across the 1,000 Monte Carlo replications,

$$\frac{1}{1000} \sum_{r=1}^{1000} |\bar{x}_{n,r} - \mu| \, .$$

Figure 1 plots this quantity. The figure is produced by simulating 1,000 samples for each sample size ranging from 1 to 1,000 coin tosses in increments of 9, calculating the sample proportion in each replication, and then computing the average absolute deviation of the sample proportion from the true probability of 0.5.

To assess the convergence rate, the empirical error path is compared with the candidate decay curves $1/\sqrt{n}$ and $1/n$. The figure shows that the average absolute error closely follows the $1/\sqrt{n}$ benchmark rather than the faster $1/n$ benchmark. This suggests that the estimation error decreases approximately in proportion to $1/\sqrt{n}$ rather than the faster rate $1/n$. In econometrics, this behavior is commonly referred to as convergence at the root-n rate. As we will see in Section 6, this square-root decay is the direct mathematical footprint of the CLT.

```
15 %% 3. Set the parameters of the simulation
16
17 % 3.1. Define alternative sample sizes (number of coin tosses)
18 N_tosses = 1:9:1000;
19
20 % 3.2. Define the number of alternative sample sizes
21 N_samples = length(N_tosses);
22
23 % 3.3. Define the number of simulated samples
24 N_sim = 1000;
25
26 %% 4. Define the population parameters
27
28 % 4.1. Define the mean for a fair coin
29 mean_population = 0.5;
30
```

```
% 4.2. Define the variance for a fair coin
var_population = 0.25;

%% 5. Generate data and calculate the sample mean

% 5.1. Preallocate memory for the simulated sample means
means_samples = NaN(N_sim,N_samples);

% 5.2. Simulate coin tosses and calculate the sample mean
for j = 1:N_samples
    % Generate N_sim independent samples of size N_tosses(j)
    sequence_of_tosses = random('Binomial',1,0.5,[N_tosses(j),N_sim]);
    % Calculate the sample mean for each simulated sample
    means_samples(:,j) = mean(sequence_of_tosses,1)';
end

%% 6. Visualizing the convergence rate

% 6.1. Calculate the average absolute distance at each sample size
mean_abs_error = mean(abs(means_samples - mean_population),1);

% 6.2. Compute the scale factor for the candidate 1/sqrt(n) decay curve
root_n_curve_scale = sqrt(N_tosses(1)) * mean_abs_error(1);

% 6.3. Generate the candidate 1/sqrt(n) decay curve
root_n_decay_curve = root_n_curve_scale ./ sqrt(N_tosses);

% 6.4. Compute the scale factor for the candidate 1/n decay curve
one_over_n_curve_scale = N_tosses(1) * mean_abs_error(1);

% 6.5. Generate the candidate 1/n decay curve
one_over_n_decay_curve = one_over_n_curve_scale ./ N_tosses;

% 6.6. Plot the empirical error path and candidate overlays
figure('Position',[1000 1000 1000 1000]);
hold on
plot(N_tosses,mean_abs_error,'.', ...
    'Color',[0.000 0.000 0.000], ...
    'DisplayName','Simulated average absolute error');
plot(N_tosses,root_n_decay_curve,'-', ...
    'Color',[0.000 0.447 0.741], ...
    'DisplayName','Candidate decay: 1/\surd{n}');
plot(N_tosses,one_over_n_decay_curve,'-', ...
    'Color',[0.301 0.745 0.933], ...
    'DisplayName','Candidate decay: 1/n');
title('Fig 1. Convergence rate of the sample proportion');
ylabel(['Mean absolute deviation of the sample proportion from ' ...
    'the true probability'])
xlabel('Sample size');
legend('show');
```

```
81 hold off
```

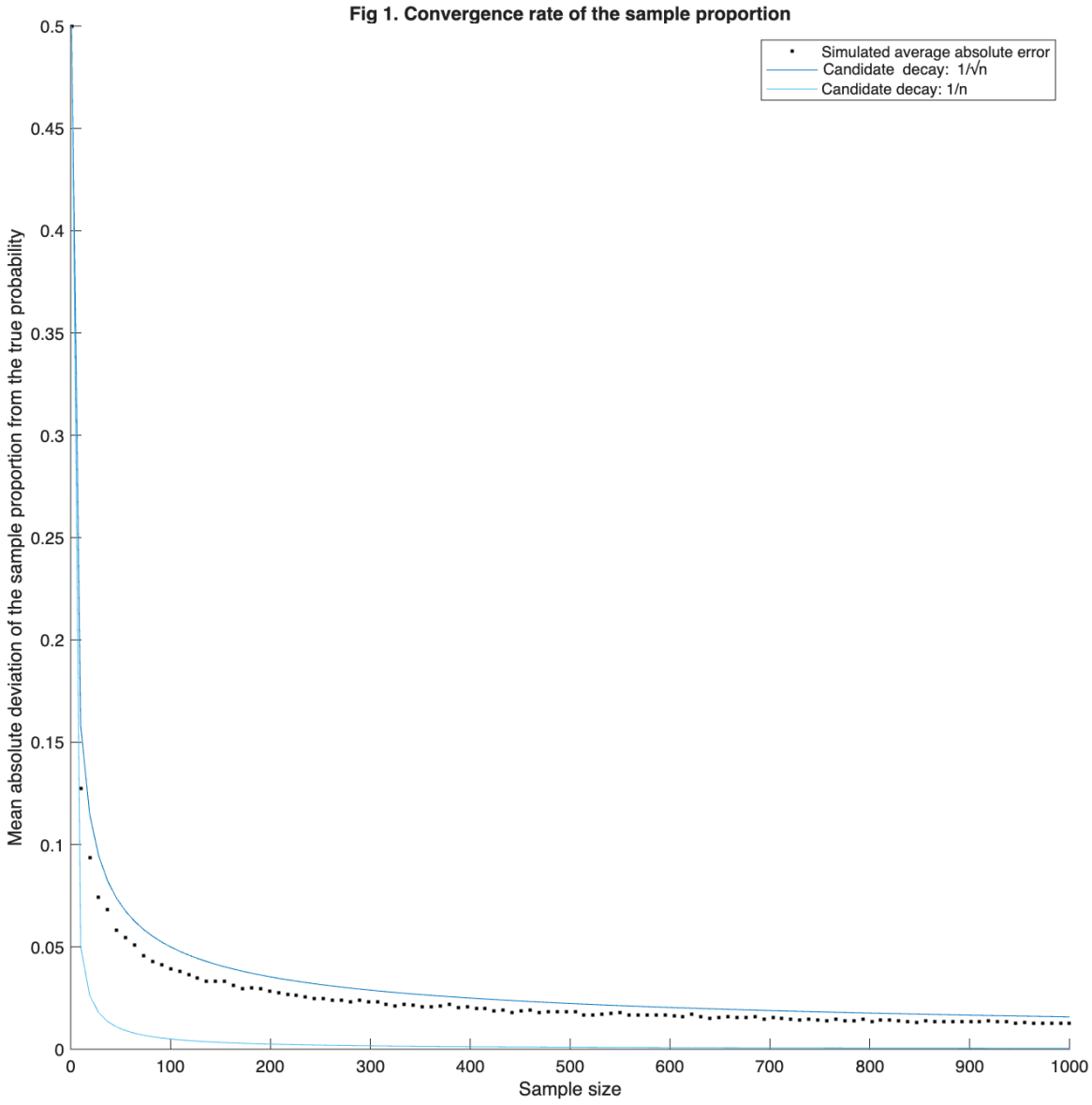


Figure 1: Comparison of the empirical error path with $1/\sqrt{n}$ and $1/n$ decay rates.

4. Convergence in probability and small o_p notation

Here we introduce the notation that describes whether a sequence of random variables vanishes in probability. This is the small-o_p notation.

A sequence of random variables x_n is said to be of order

$$o_p(a_n)$$

if, for any constant $\delta > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{x_n}{a_n}\right| \geq \delta\right) = 0.$$

In Section 2, we showed through the Weak Law of Large Numbers that the sample mean $\bar{x}_n$ converges in probability to the population mean μ:

$$\lim_{n\to\infty} \mathbb{P}(|\bar{x}_n - \mu| < \varepsilon) = 1.$$

Equivalently, the probability that the estimation error exceeds any fixed threshold approaches zero as the sample size increases:

$$\lim_{n\to\infty} \mathbb{P}\left(|\bar{x}_n - \mu| \geq \delta\right) = 0.$$

Applying the definition with

$$x_n = \bar{x}_n - \mu$$

and

$$a_n = 1,$$

gives

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{\bar{x}_n - \mu}{1}\right| \geq \delta\right) = 0.$$

Therefore,

$$\bar{x}_n - \mu = o_p(1).$$

The notation $o_p(1)$ indicates that the estimation error converges in probability to zero.

Figure 2 illustrates convergence in probability using the coin-toss example. We select four representative sample sizes and, for each of them, plot the Monte Carlo realizations of the unscaled estimation error, $\bar{x}_n - \mu$. To provide a theoretical benchmark, we overlay a 95% decay envelope given by $\pm 1.96\sigma/\sqrt{n}$, which shrinks toward zero as the sample size increases. The resulting figure shows that the dispersion of the unscaled estimation error collapses as the sample size increases. This visual pattern is consistent with the Weak Law of Large Numbers and illustrates the meaning of small-o_p notation, namely that the estimation error converges in probability to zero.

```
%% 7. Select representative sample sizes for visualization

% 7.1. Define representative fractions of the sample-size grid
fractions = [0.10, 0.30, 0.50, 1.00];

% 7.2. Locate the corresponding positions in the sample-size grid
target_indices = max(1,round(fractions * N_samples));
```

```
90
91  % 7.3. Extract the corresponding sample sizes
92  target_n = N_tosses(target_indices);
93
94  %% 8. Visualizing convergence in probability: op(1)
95
96  % Plot the unscaled estimation errors
97  figure('Position',[1000 1000 1000 1000]);
98  hold on
99  for k = 1:length(target_indices)
100     % The column index in means_samples
101     j = target_indices(k);
102     % The exact sample size for this iteration
103     plot_n = N_tosses(j);
104     % Compute the unscaled estimation error
105     unscaled_error_col = means_samples(:,j) - mean_population;
106     % Replicate the sample size value for all Monte Carlo simulations
107     x_axis_values = repmat(plot_n,N_sim,1);
108     % Plot the unscaled estimation errors for this sample size
109     scatter(x_axis_values,unscaled_error_col,5,'.', ...
110         'MarkerEdgeColor',[0.000 0.000 0.000], ...
111         'MarkerEdgeAlpha',0.25, ...
112         'HandleVisibility','off');
113 end
114 % Construct the theoretical 95% decay envelope
115 n_continuous = linspace(min(N_tosses),max(N_tosses)*1.05,500);
116 theoretical_bound = ...
117     (1.96 * sqrt(var_population)) ./ sqrt(n_continuous);
118 % Overlay the theoretical decay envelope
119 plot(n_continuous,theoretical_bound,'-', ...
120     'Color',[0.850 0.325 0.098]);
121 plot(n_continuous,-theoretical_bound,'-', ...
122     'Color',[0.850 0.325 0.098], ...
123     'HandleVisibility','off');
124 % Add a reference line at zero estimation error
125 yline(0,'-', ...
126     'Color',[0.000 0.000 0.000], ...
127     'HandleVisibility','off');
128 title('Fig 2. Convergence in probability of the unscaled error');
129 ylabel(['Unscaled deviation of the sample proportion from the true ' ...
130     'probability']);
131 xlabel('Sample size');
132 % Dynamically set the x-axis limit based on the maximum sample size
133 xlim([0,max(N_tosses)*1.05]);
134 % Display only the selected sample sizes on the x-axis
135 xticks(target_n);
136 hold off
```

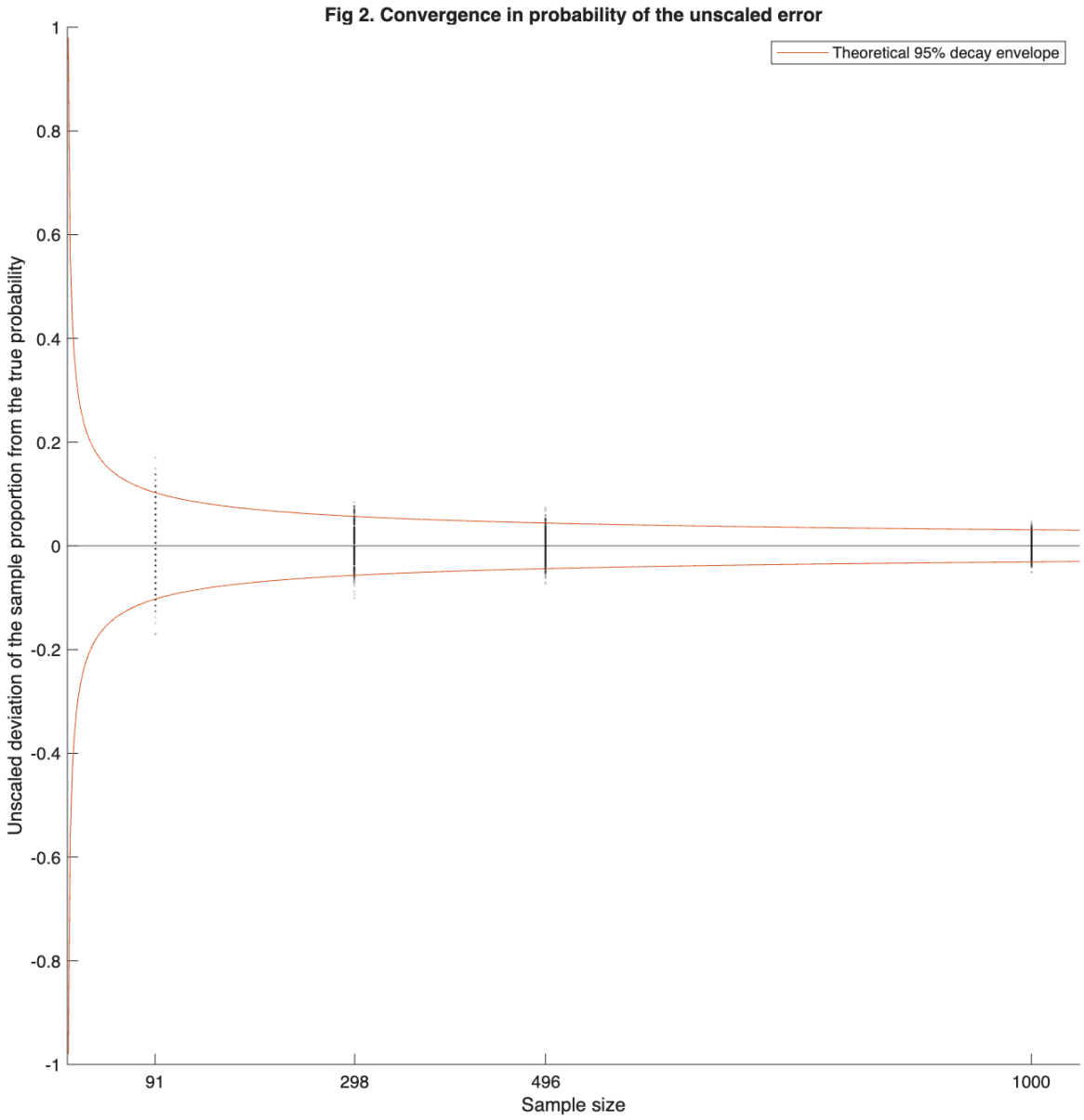


Figure 2: Illustration of convergence in probability.

5. Stochastic boundedness and Big $\mathcal{O}_p$ notation

Here we introduce the notation that describes stochastic boundedness. This is the Big-$\mathcal{O}_p$ notation. A sequence of random variables x_n is said to be at most of order

$$\mathcal{O}_p(a_n)$$

if, for any arbitrarily small probability $\varepsilon > 0$, there exists a finite constant M and an integer N such that

$$\mathbb{P}\left(\left|\frac{x_n}{a_n}\right| > M\right) < \varepsilon, \qquad \text{for all } n > N.$$

This condition states that the sequence, when scaled by a_n, remains bounded in probability.

Using the same representative sample sizes as in Figure 2, we compute the root-n scaled estimation error,

$$\sqrt{n}(\bar{x}_n - \mu),$$

for each Monte Carlo replication and plot the resulting observations against the corresponding sample sizes in Figure 3.

Unlike the funneling effect observed in Figure 2, the vertical spread of the scaled errors does not collapse toward zero. Instead, the cloud of observations stabilizes within a roughly constant horizontal band. As the sample size increases, the observations neither concentrate at a single point nor spread out toward infinity.

This visual pattern is consistent with the definition of $\mathcal{O}_p(1)$. The root-n scaled estimation error remains confined within roughly fixed bounds, suggesting that the sequence is stochastically bounded. The roughly constant vertical spread of the cloud reflects the finite constant M appearing in the definition of Big-$\mathcal{O}_p$ notation.

```
%% 9. Visualizing stochastic boundedness: Op(1)

% Plot the root-n scaled estimation errors
figure('Position',[1000 1000 1000 1000]);
hold on
for k = 1:length(target_indices)
    % The column index in means_samples
    j = target_indices(k);
    % The exact sample size for this iteration
    plot_n = N_tosses(j);
    % Compute the root-n scaled estimation error
    scaled_error_col = sqrt(plot_n) * ...
        (means_samples(:,j) - mean_population);
    % Replicate the sample size value for all Monte Carlo simulations
    x_axis_values = repmat(plot_n,N_sim,1);
    % Plot the root-n scaled estimation errors for this sample size
    scatter(x_axis_values,scaled_error_col,5,'.', ...
    'MarkerEdgeColor',[0.000 0.000 0.000], ...
    'MarkerEdgeAlpha',0.25);
end
% Reference line corresponding to zero estimation error
yline(0,'-', ...
    'Color',[0.850 0.325 0.098]);
```

```
title('Fig 3. Stochastic boundedness of the root-n scaled error');
ylabel(['Root-n scaled deviation of the sample proportion from ' ...
    'the true probability'])
xlabel('Sample size');
% Dynamically set the x-axis limit based on the actual max value
xlim([0,max(N_tosses) * 1.05]);
% Fix the y-axis range
ylim([-2,2]);
% Display only the selected sample sizes on the x-axis
xticks(target_n);
hold off
```

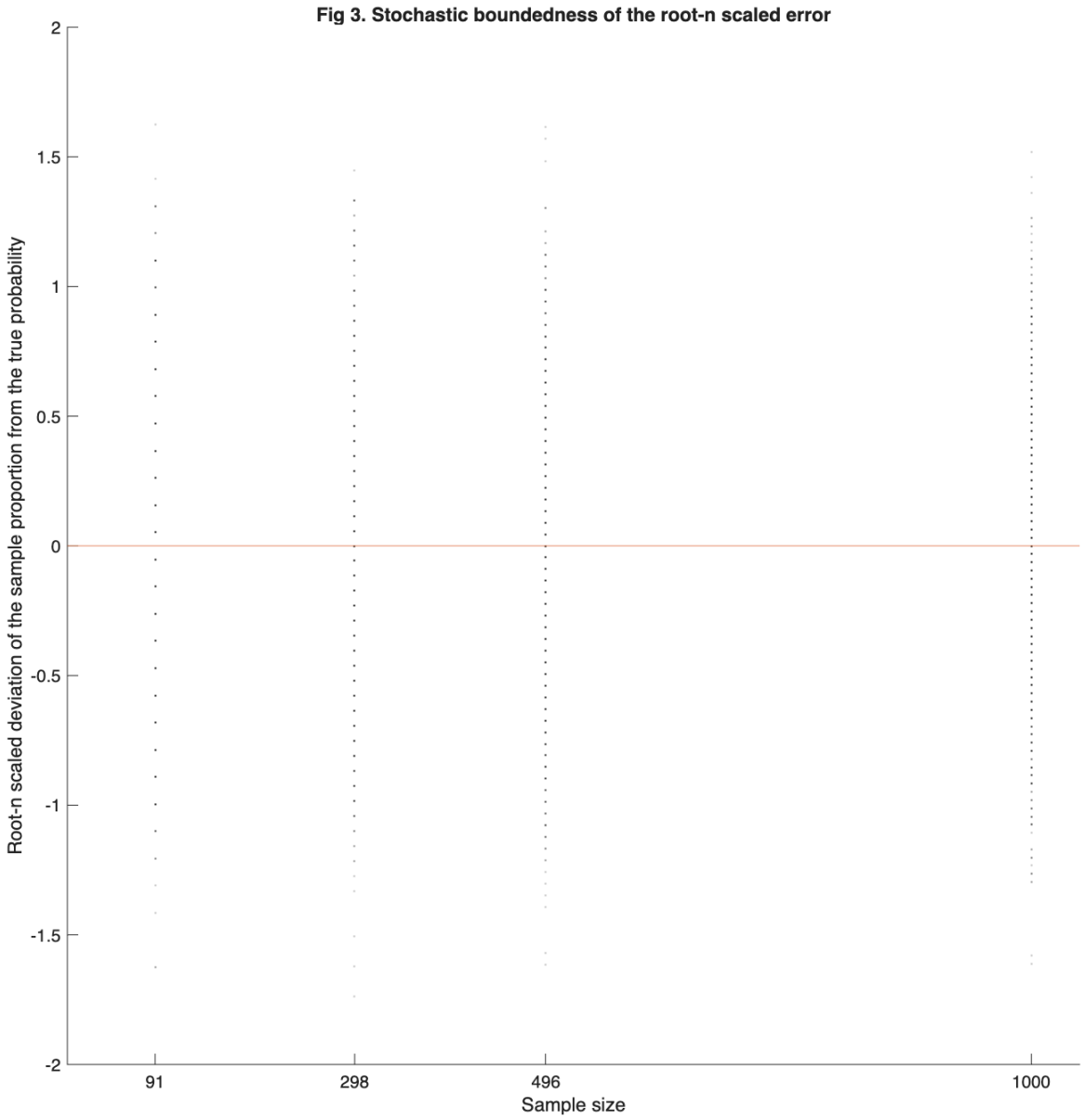


Figure 3: Stochastic boundedness of the root-n scaled estimation error.

6. Linking Big $\mathcal{O}_p$ to the central limit theorem

Figure 3 showed that the root-n scaled estimation error,

$$\sqrt{n}\,(\bar{x}_n - \mu)\,,$$

remains confined within a roughly constant band as the sample size increases. The CLT provides a theoretical explanation for this behavior.

Under the assumption that the data are i.i.d. with mean μ and finite variance σ^2,

$$\sqrt{n}\,(\bar{x}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Unlike the convergence in probability studied earlier, this result does not imply that the scaled estimation error converges to zero. Instead, it states that the distribution of the scaled estimation error approaches a stable normal distribution.

To visualize this result, we estimate the empirical probability density of the scaled error

$$\sqrt{n}\,(\bar{x}_n - \mu)$$

for the largest simulated sample size, 1,000. We compare this empirical density with the theoretical density of a Normal distribution with mean 0 and variance $\sigma^2 = 0.25$. The result is presented in Figure 4.

The empirical density closely mirrors the shape of the theoretical Normal curve. This overlap provides visual evidence for the CLT. Whereas Figure 2 illustrated convergence toward a single point, Figure 4 illustrates convergence toward an entire probability distribution.

This limiting Normal distribution also explains the stochastic boundedness observed in Figure 3. Because the root-n scaled estimation error converges to a distribution with finite variance, it does not spread out toward infinity. Consequently,

$$\sqrt{n}\,(\bar{x}_n - \mu) = \mathcal{O}_p(1).$$

Equivalently,

$$\bar{x}_n - \mu = \mathcal{O}_p\left(\frac{1}{\sqrt{n}}\right).$$

Hence, the CLT provides the theoretical justification for the root-n convergence rate observed in Figure 1.

```
%% 10. Visualizing asymptotic normality of the root-n scaled error

% 10.1. Select the largest sample size for the CLT demonstration
n_large = N_tosses(N_samples);

% 10.2. Calculate the root-n scaled estimation error
scaled_error_clt = sqrt(n_large) * ...
    (means_samples(:,N_samples) - mean_population);

% 10.3. Estimate the empirical density of the scaled errors
[f_empirical,x_empirical] = ksdensity(scaled_error_clt);

% 10.4. Calculate the theoretical standard deviation implied by the CLT
```

```
std_dev = sqrt(var_population);

% 10.5. Create an evaluation grid for the theoretical Normal density
x_theoretical = linspace(min(x_empirical),max(x_empirical),500);

% 10.6. Evaluate the theoretical Normal density on the grid
f_theoretical = normpdf(x_theoretical,0,std_dev);

% 10.7. Compare the empirical and theoretical densities
figure('Position',[1000 1000 1000 1000]);
hold on
% Plot the empirical kernel density estimate
plot(x_empirical,f_empirical,'-', ...
    'Color',[0.000 0.000 0.000], ...
    'DisplayName','Empirical density of the scaled error');
% Overlay the theoretical Normal density
plot(x_theoretical,f_theoretical,'-', ...
    'Color',[0.850 0.325 0.098], ...
    'DisplayName','Theoretical Normal density');
title('Fig 4. Asymptotic normality of the root-n scaled error');
xlabel(['Root-n scaled deviation of the sample proportion from ' ...
    'the true probability']);
ylabel('Probability density');
legend('show');
hold off
```

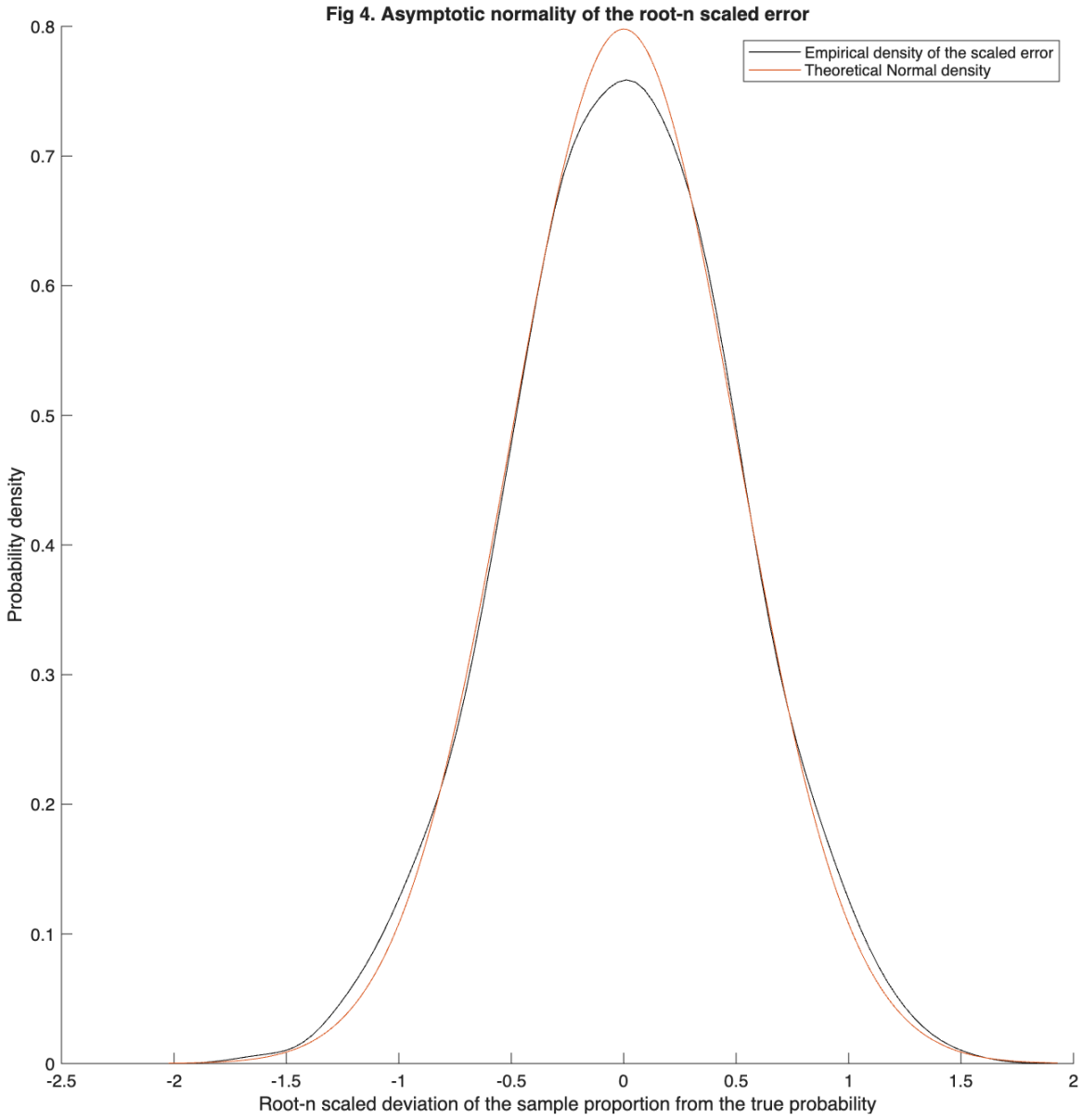


Figure 4: Comparison of the empirical density and the limiting Normal density.

7. Final notes

This file is prepared and copyrighted by Robin van der Meulen and Tunga Kantarcı. This file and the accompanying MATLAB file are available on GitHub and can be accessed using this link.